

Hints and Solutions :

1. According to Faraday's law, magnetic field changing with time gives rise to an electric field and the fact that an electric field changing with time gives rise to a magnetic field, is the symmetrical counterpart, and is a consequence of the displacement current being a source of magnetic field. Thus, time dependent electric and magnetic field give rise to each other

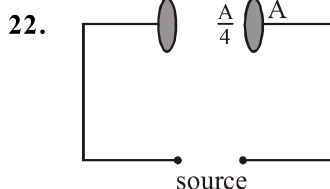
$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$$

2. NCERT Pg.#281
 3. Infrared rays are sometimes referred as heat waves. (NCERT Pg.#282)
 4. NCERT Pg.#282
 5. NCERT Pg.#282
 6. NCERT Pg.#282
 7. NCERT Pg.#283
 8. NCERT Pg.#283
 9. NCERT Pg.#274
 10. NCERT Pg.#277
 14. In a wave equation, variation can be in pressure, electric and magnetic field.
 15. The heating effect produced by infrared radiations can be detected by pyrometer.
 17. All EM waves travel in vacuum with the same speed c .

$$8. I = \frac{P}{A} = \frac{3.9 \times 10^{26}}{4\pi r^2} = 6.4 \times 10^7$$

$$0. Z = \sqrt{\frac{\mu_r}{\epsilon_r}} \times \sqrt{\frac{\mu_0}{\epsilon_0}} \\ = \sqrt{\frac{50}{2}} \times 376.6 \\ = 1883 \Omega$$

1. When capacitor is fully charged then EF inside it become constant and there will be no induced magnetic field.



$$I_d = I_e - I$$

$$\text{Now current density } J = \frac{I}{A} \quad \left| \quad \left(I'_d \right) = J \times \frac{A}{4} \right.$$

$$\text{So, current through } \frac{A}{4} \text{ area will be } \quad \left| \quad I'_d = \frac{I}{4} \right.$$

23. Frequency of rotation of water molecules is 2.45 GHz. When microwave of this frequency incident over water molecules then water molecules absorb this radiation which is equivalent to heating of water. It is basic concept of working of microwave oven.
 24. Due to large molecules size vibrational and rotational frequencies are much smaller to absorb energy of radiations.
 25. If an AM radio has a telescopic antenna it responds to electric part of the signal when antenna is turned horizontal then signals are greatly diminished.
 26. Water molecules present in most materials readily absorb IR radiation. After absorption their thermal motion increases i.e., they heat up and heat then surrounding.
 27. Semiconductor devices like LED etc. emit IR radiations which is used in TV remote.
 28. UV radiations can be focussed into very narrow beams for high precision application such as LASIK eye surgery.
 29. EM waves carry energy which is shared equally in electric and magnetic fields. It transport momentum also as $p = \frac{U}{C}$ where U is energy transferred to surface in time ' t '.
 30. Klystron valve, thermionic electron tube generates or amplifies microwaves by controlling the speed of a stream of electrons.
 31. A Geiger counter (Geiger-Muller tube) is a device used for the detection and measurement of all types of radiations α , β and γ radiations.